Experiment: 8
Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim

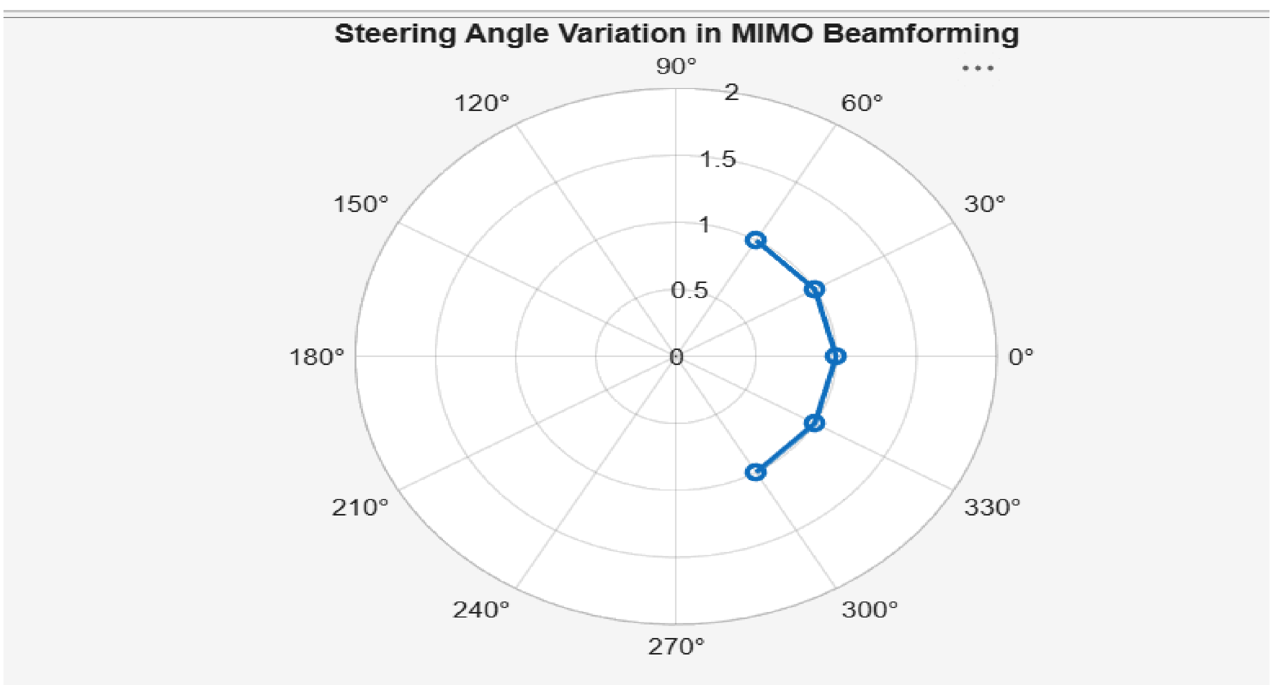
To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objectives

1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.
2. To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.
3. To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

Objective – 1 Analyze Steering Angle (Degrees)

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60];
gain = ones(size(angle));
figure
% Steering Angles (degrees)
% Normalized Gain
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)
title('Steering Angle Variation in MIMO Beamforming')
```



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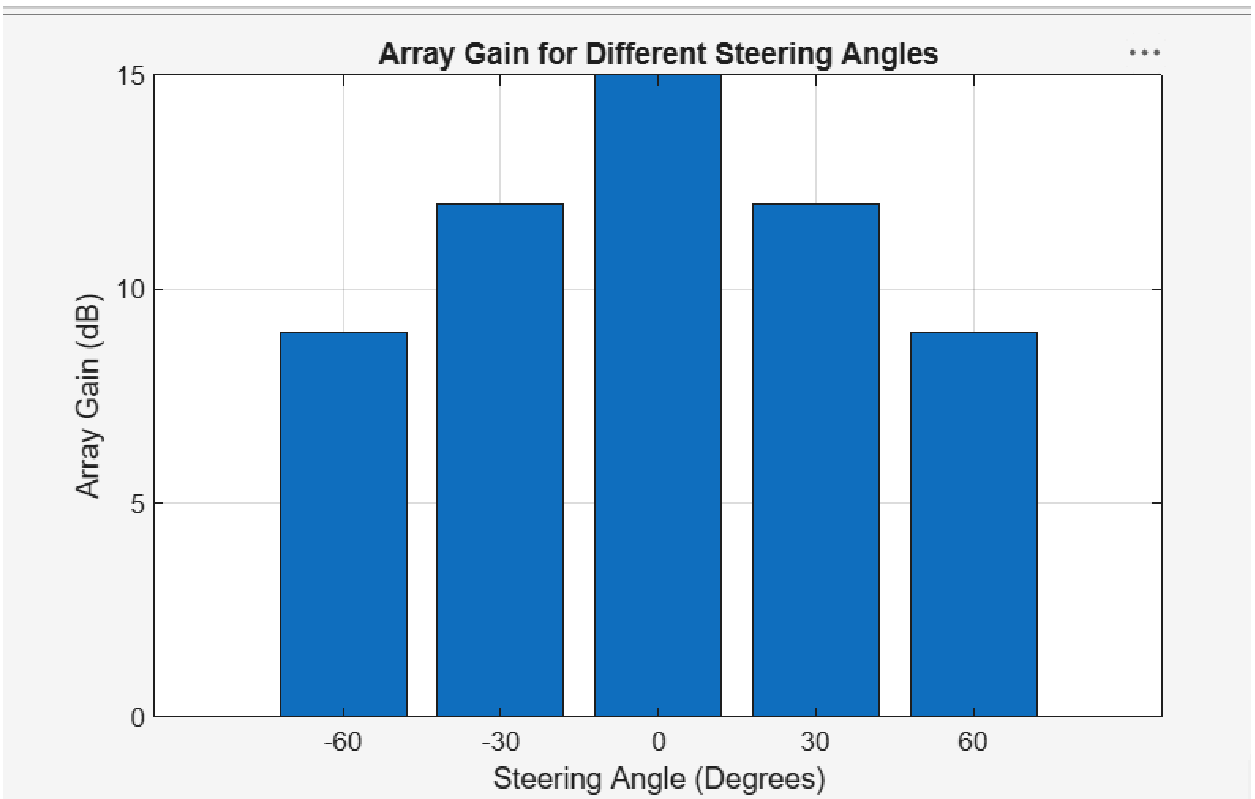
Objective – 2 Compare Array Gain (dB)

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60];
gain = [9 12 15 12 9];
figure
bar(angle, gain)
grid on
% Steering Angles (degrees)
% Array Gain (dB)
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')
```



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Objective – 3 Evaluate Beam Direction Error (Degrees)

```
clc;
clear;
close all;
desired = [-60 -30 0 30 60];
actual = [-58 -32 1 29 62];
error = abs(desired - actual);
figure
plot(desired, error, 'o-', 'LineWidth', 2)
grid on
xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')
```